6.6 Solving Exponential and Logarithmic Equations

Name _____ Period ____ Date _____

Algebra II 6.6 (30-minute plan) · Student sheet

I can…

- solve exponential equations.
- solve logarithmic equations and check for extraneous solutions.
- solve exponential inequalities and real-life problems.

Vocabulary · fill in the missing word

Property of Equality for Exponential Equations — $b^x =$ _____ $= y$

Property of Equality for Logarithmic Equations — $\log_b x =$ _____ $= y$

Extraneous solution — an answer that makes the log of a _____ number in the original equation

Part 1 · Notes and examples (fill in each note, then work the example)

Same base: If $b^x = b^y$, then _____.

Example 1

Solve $9^x = 27^{x-1}$.

Take a log: If the bases can't match, take a _____ of both sides: $b^x = a \rightarrow x =$ _____.

Example 2

Solve $4^x = 11$. Round to three decimal places.

Logarithmic equations: To solve $\log_b x = c$, rewrite it as _____.

Check: Check every answer: the inside of a logarithm must be _____.

Example 3

Solve $\ln(4x - 7) = \ln(x + 5)$.

Part 2 · Partner practice

Partner practice

Solve $5^x = 30$. Round to three decimal places.